STACKS, SHEAVES ON ALGEBRAIC STACKS: LEMMA-QUASI-COHERENT-ALTERNATIVE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 4750-4773 of sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex; proof span is lines 4775-4796, the complete author proof (20 body lines).

Editorial note. Navigation stubs: 7 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 20 of queue_algebraic-stacks-sheaves.json (pb=1260, stmt 4750-4773, proof 4775-4796).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma definition-alternative: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 4639–4646. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0GZV: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 4796–4796. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma equation-alternative-ringed: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 4727–4731. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma section-restriction: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 1185–1200. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 03DL: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 4796–4796. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0GZN: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 4796–4796. Cited by the retained statement or

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proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 03DG: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 4796–4796. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $p : \mathcal{X} \to (Sch/S)_{fppf}$ be a category fibred in groupoids. Let $\mathcal{F}$ be an $\mathcal{O}$-module on $\mathcal{X}_{affine}$. The following are equivalent*

1. *for every morphism $x \to x'$ of $\mathcal{X}_{affine}$ the map $\mathcal{F}(x') \otimes_{\mathcal{O}(x')} \mathcal{O}(x) \to \mathcal{F}(x)$ is an isomorphism,*
2. *$\mathcal{F}$ is a quasi-coherent module on $(\mathcal{X}_{affine}, \mathcal{O})$ in the sense of Modules on Sites, Definition 03DL,*
3. *$\mathcal{F}$ is a sheaf for the Zariski topology on $\mathcal{X}_{affine}$ and a quasi-coherent module on $(\mathcal{X}_{affine,Zar}, \mathcal{O})$ in the sense of Modules on Sites, Definition 03DL,*
4. *same as in (3) for the étale topology,*
5. *same as in (3) for the smooth topology,*
6. *same as in (3) for the syntomic topology,*
7. *same as in (3) for the fppf topology, and*
8. *$\mathcal{F}$ corresponds to a quasi-coherent module on $\mathcal{X}$ via the equivalence (equation-alternative-ringed).*

Proof. To make sense out of part (2), recall that $\mathcal{X}_{affine}$ is a site gotten by endowing the category $\mathcal{X}_{affine}$ with the chaotic topology (Definition definition-alternative) and hence a sheaf of $\mathcal{O}$-modules $\mathcal{F}$ is the same thing as a presheaf of $\mathcal{O}$-modules. Conditions (1) and (2) are equivalent by Modules on Sites, Lemma 0GZN. Observe that for $\tau \in \{Zariski, étale, smooth, syntomic, fppf\}$ the presheaf $\mathcal{F}$ is a τ-sheaf if and only if for all $x \in \mathrm{Ob}(\mathcal{X}_{affine})$ the restriction to $\mathcal{X}_{affine}/x$ is a τ-sheaf. Set $U = p(x)$. Similarly to the discussion in Section section-restriction the object x of $\mathcal{X}_{affine}$ induces an equivalence $\mathcal{X}_{affine,étale}/x \to (Aff/U)_{étale}$ of sites. In this way we see that the equivalence of (1) with (3) – (7) follows from Descent, Lemma 0GZV applied to each of these sites. The equivalence of (8) and (7) is immediate from the fact that "being quasi-coherent" is an intrinsic property of sheaves of modules, see Modules on Sites, Section 03DG □